Homework 3

Due September 16, 2025.

Problem 1. Q_{2n}

Consider the dicyclic group Q_{2n} whose presentation is as follows

$$Q_{2n} = \{a, b | a^{2n} = e, a^n = b^2, bab^{-1} = a^{-1}\} \tag{1}$$

In class we discussed the quaternion group Q_4 of order 8; now we consider Q_6 of order 12.

a) What are the group elements of Q_6? What is their order? Do all group elements in a given class have the same order? If $ba^k = a^l b$ what are k and l? What are the classes of Q_6? Draw a circle around each class, and write the order of each class under the class.

b) Q_6 has 3 proper normal subgroups; what are they? What is the commutator subgroup $C(Q_6)$?

c) Show that Q_6 can be written as the semidirect product $Z_3 \rtimes Z_4$. (*Hint:* $Z_4 = \{e, b, b^2, b^3\}$.)

d) What is the center $Z(Q_6)$ of Q_6? What is the centralizer Z_b of b?

e) Prove that the following is a representation (a realization in terms of matrices) of the group Q_6

$$a = \tfrac{1}{2}(I + i\sqrt{3}\sigma_3)\,; \qquad b = i\sigma_2 \quad (\text{or } b = i\sigma_1), \tag{2}$$

where I is the 2×2 unit matrix and σ_i are the Pauli matrices.

f) As we shall prove in class, there are as many irreducible representations of a finite group as there are classes. If d_{R_i} denotes the dimension of the representation R_i, then $|G| = \sum_{i=1}^{n_C} \left(d_{R_i}\right)^2$ where n_C is the number of classes. The number of 1-dimensional representation is $|G|/C(G)$. Use these theorems to deduce the dimensions of the irreducible representation of Q_6.

Problem 2. A_5

The group A_5 is famous for two reasons: it is the first simple group after the cyclic groups of prime order, and it is the rotation group of an icosahedron. We shall study both properties.

a) What is the order of A_5? What are the classes of A_5? (Recall that the 5-cycles form two classes.) What is the order of these classes? If A_5 were not simple, it would have a proper normal subgroup. A proper normal subgroup consists of entire classes. Show that no combination of classes yields a subgroup. (Use that the order of a subgroup is a divisor of the order of the group.) This proves that A_5 is simple. [It is claimed that Evariste Galois was the first to prove that A_5 is simple. (His proof was lost or never recorded.) He used this property for his proof that not all quintic polynomial equations have only radical roots.]

b) Consider an icosahedron. Show that the number of vertices V, edges E, and faces F, satisfies Euler's equation $V - F + E = 2$. How many rotations are there around an axis that connects two opposite vertices? And about an axis through the middle of two opposite faces? (We call such as axis a crossline.) And through the middle of two opposite edges?

c) Now we are going to show that these rotations correspond to the permutations of the alternating group A_5. To show this we consider the 15 crosslines and divide them into 5 triplets. A triplet is constructed as follows. Put an icosahedron in front of you on a table such that it is resting on an edge (you must keep it upright). The vertical crossline is the first of the triplet we will construct. Next consider a horizontal plane through the center of the icosahedron. There are two crosslines in this plane; together with the vertical crossline this constitutes one of the five triplets. Draw a picture of the intersection of the horizontal plane with the icosahedron, and draw in this picture the two orthogonal crosslines. [We will put on reserve in the Math-Phys library the icosahedron shown in class. You can ask for it to study it, but you cannot check it out.]

d) Each rotation corresponds to an even permutation of the five triplets. Identify the rotations established before with the group elements of A_5. This proves that A_5 is the rotation group of A_5.

e) What is the full isometry group of an icosahedron (rotations and reflections)? Is it S_5? Or $A_5 \times (e, i)$ where i is space inversion? Or something else? [Felix Klein, of the Kelin group V, studied the icosahedron, in his "Erlanger program" of 1872 in which he proposed to classify and unify different geometries by focussing on their group properties.]